Module 4: Activation Functions and Vanishing Gradients

1. What Is an Activation Function?

An activation function transforms the weighted sum calculated by a neuron into its output.

Recall the equation of an artificial neuron:

$z = Wx + b$

The activation function is applied to this value:

$a = f(z)$

Where:

- z = weighted sum before activation
- f = activation function
- a = output of the neuron

Activation functions introduce non-linearity into neural networks.

2. Why Do We Need Activation Functions?

Without non-linear activation functions, stacking multiple dense layers still produces an overall linear transformation.

This limits the kinds of relationships the network can learn.

Non-linear activation functions allow a network to model more complex patterns in data.

For example, identifying a cat from image pixels requires learning complex relationships between different visual features.

3. Types of Activation Functions

3.1 Linear Activation Function

Formula:

$f(x) = x$

Output range: All real numbers.

The output is equal to the input.

Example:

If x = 5, then f(x) = 5.

Uses:

- Commonly used in the output layer for regression.
- Usually not useful in hidden layers of a standard feed-forward network because it does not introduce non-linearity.

3.2 Sigmoid Activation Function

Formula:

$\sigma(x) = 1 / (1 + e^{-x})$

Output range: Between 0 and 1.

The sigmoid function converts a real-valued input into a value between 0 and 1.

Example:

If x = 0:

$\sigma(0) = 1 / (1 + 1)$

$\sigma(0) = 0.5$

Uses:

- Commonly used in the output layer for binary classification.
- Can represent a probability-like prediction for a binary target when trained appropriately.

Limitations:

For very large positive or negative inputs, sigmoid saturates. Its derivative becomes very small in these regions, which can contribute to the vanishing gradient problem.

3.3 Tanh Activation Function

Formula:

$\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$

Output range: Between −1 and 1.

Example:

$\tanh(0) = 0$

Unlike sigmoid, tanh is zero-centered.

Uses:

- Has been widely used in neural networks, including recurrent neural networks.
- Can be useful when negative and positive activations are desirable.

Limitations:

Like sigmoid, tanh saturates for very large positive or negative inputs. Its derivatives can become very small, contributing to vanishing gradients.

3.4 ReLU Activation Function

ReLU stands for Rectified Linear Unit.

Formula:

$\text{ReLU}(x) = \max(0, x)$

This means:

- If $x < 0$, output = 0.

- If x > 0, output = x.
- If x = 0, output = 0.

Examples:

ReLU(-5) = 0

ReLU(0) = 0

ReLU(7) = 7

Uses:

- Commonly used in hidden layers of feed-forward neural networks.
- Widely used in convolutional neural networks.
- Helps reduce saturation-related gradient problems for positive inputs.

Limitations:

Some neurons can become inactive if they consistently receive negative inputs. This is commonly called the *dying ReLU problem*.

4. Activation Function Comparison

Function	Output Range	Common Use	Main Limitation
Linear	Any real number	Regression output	No non-linearity
Sigmoid	(0, 1)	Binary classification output	Can cause vanishing gradients
Tanh	(-1, 1)	Some hidden layers and recurrent networks	Can cause vanishing gradients

ReLU	[0, ∞)	Hidden layers	Can suffer from dying neurons

5. Understanding the Vanishing Gradient Problem

The vanishing gradient problem occurs when gradients become extremely small as they are propagated backward through a deep neural network.

As a result, earlier layers may receive gradients that are too small to produce meaningful parameter updates.

These layers can learn very slowly or stop learning effectively.

Why Does This Happen?

During backpropagation, the chain rule combines derivatives across multiple layers.

When many of these derivatives have small magnitudes, multiplying them together can produce an extremely small gradient.

For example:

$0.1 \times 0.1 \times 0.1 \times 0.1 = 0.0001$

With many layers, this effect can become more severe.

This example illustrates the multiplication of small values; it is not the exact gradient calculation for every network.

6. Why Can Sigmoid and Tanh Contribute to Vanishing Gradients?

Sigmoid and tanh both saturate at extreme input values.

In their saturated regions, their derivatives are close to zero.

During backpropagation, these small derivatives can cause gradients to shrink as they travel through multiple layers.

Important Correction

The vanishing gradient problem is not simply that the neuron's output becomes smaller and smaller until it reaches zero.

The central problem is that **the gradients used to update the parameters become extremely small**.

Consequently, the parameters of earlier layers may not learn effectively.

7. How Can We Reduce Vanishing Gradients?

Several techniques can help:

1. **ReLU and related activations:** These avoid sigmoid-like saturation on their positive regions.
2. **Suitable weight initialization:** Methods such as He initialization can help maintain useful signal and gradient scales in ReLU networks.
3. **Normalization:** Techniques such as batch normalization can improve training behavior in suitable architectures.
4. **Residual connections:** Skip connections help gradients flow through very deep networks.
5. **Appropriate architecture:** The network structure and activation choices should suit the problem.

No single technique guarantees that vanishing gradients will never occur.

8. Which Activation Function Should We Use?

The appropriate function depends on the task and the layer.

Hidden Layers

ReLU is a common starting point for standard feed-forward networks and many CNNs.

Other choices include Leaky ReLU, GELU, and SiLU, depending on the architecture.

Binary Classification Output

A sigmoid output is a common choice when predicting one binary target.

Multi-Class Classification Output

Softmax is commonly used to convert class scores into a distribution of probabilities when exactly one class is expected.

Regression Output

A linear output is common when the target can take unrestricted real values.

9. Quick Revision Questions

1. What is an activation function?
2. Why do neural networks need non-linearity?
3. What is the output range of sigmoid?
4. What is the difference between sigmoid and tanh?
5. How does ReLU work?
6. What is the vanishing gradient problem?
7. Why can sigmoid and tanh contribute to vanishing gradients?
8. Why is ReLU commonly used in hidden layers?
9. Which output activation is commonly used for binary classification?
10. Which output activation is commonly used for unrestricted regression?

Module Summary

Activation functions introduce non-linearity into neural networks. Linear, sigmoid, tanh, and ReLU have different output ranges and behaviors. Sigmoid and tanh can contribute to vanishing gradients in deep networks because their derivatives become very small in saturated regions. ReLU is a common hidden-layer choice, while the output activation should match the task.